F79SP Session 18: Results on probability theory

1 Probability Theory

We will review relevant probability theories for Poisson Process. Proof can be found in notes.

Poisson distribution

- Property Π2: Let $N_1 \sim \Pi(\lambda_1)$ and $N_2 \sim \Pi(\lambda_2)$, and N1 and N2 are independent. Then, for any $k = 0, 1, 2, ...$:

$$N_1 \,|\, N_1 + N_2 = k \text{ has Binomial distribution } B\left(k, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

- Property Π2′: Generalise to more than two Poisson distributions!
- Property Π3: Let $N \sim \Pi(\lambda)$ for $\lambda > 0$ be the number of balls. Independent of everything else, color each ball with red (with probability p) or green (with probability $q = 1 - p$). Let R and G be the total number of red balls and green balls respectively. Then:
  1. $R + G = N$
  2. $R \sim \Pi(\lambda p)$
  3. $G \sim \Pi(\lambda q)$
  4. R and G are independent.

Geometric distribution

- Property G1: (Memoryless property) If $X \sim G(p)$, then, for any $k, m = 0, 1, \ldots$:

$$\mathbf{P}(X > k + m \,|\, X > k) = \mathbf{P}(X > m).$$

Exponential distribution

- Renyi Theorem: Let $p \to 0$. Then $pG(p) \to Exp(1)$.
- Property E1: (Memoryless property) If $Y \sim Exp(\alpha)$, then for any $x, y \geq 0$,

$$P(Y > x + y \,|\, Y > x) = P(Y > y).$$

- Property E4: Let $Y_1 \sim Exp(\alpha_1)$ and $Y_1 \sim Exp(\alpha_2)$. Then $\min(Y_1, Y_2) \sim Exp(\alpha_1 + \alpha_2)$

Uniform distribution

- Property U3: Let $X_1, X_2, \ldots, X_n$ be i.i.d. r.v.'s with $U(a, b)$ distribution. For $a < c < b$, let $Z = |\{i : X_i \in (a, c)\}|$, then $Z \sim B(n, \frac{c-a}{b-a})$.